

8/18/2026 Lecture

I. Soft actor-critic

A. Preliminaries

1. $p_{\pi}(s_k), p_{\pi}(s_k, a_k)$ are laws induced by policy $\pi(a_k | s_k)$

2. I hate this notation

2. objective:

$$J(\pi) = \mathbb{E}_{t=0}^{\infty} \mathbb{E}_{(s_t, a_t) \sim p_{\pi}} (r(s_t, a_t) + \gamma V_{\pi}(s_{t+1}))$$

note: NO discount factor!

B. From Soft Policy Iteration to Soft Actor-Critic

1. modified Bellman backup operator:

$$T^{\pi} Q(s_k, a_k) = r(s_k, a_k) + \gamma \mathbb{E}_{s_{k+1} \sim p} (V(s_{k+1}))$$

no 1

$$\text{exactly... } 1. \text{ for } V(s_k) = \mathbb{E}_{a \sim \pi} (Q(s_k, a) - \log \pi(a | s_k))$$

2. goal: fixed point of this eq'n

3. Π = set of parameterized policies

4. policy update:

$$\pi_{\text{new}} = \arg \min_{\pi \in \Pi} D_{KL}(\pi(\cdot | s_k) \| \frac{\exp(Q^{\pi_{\text{old}}}(s_k, \cdot))}{Z^{\pi_{\text{old}}}(s_k)})$$

5. now, for ϕ as a space, need parameters

$V_{\phi}, Q_{\phi}, \pi_{\phi}$

6. train V_{ϕ} to minimize

$$J_V(\phi) = \mathbb{E}_{s \sim \rho} (\frac{1}{2} (V_{\phi}(s) - \mathbb{E}_{a \sim \pi_{\phi}} (Q_{\phi}(s, a) - \log \pi_{\phi}(a | s)))^2)$$

a. one point estimator:

$$\hat{S}_{\phi} J_V(\phi) = \nabla_{\phi} V_{\phi}(s) (V_{\phi}(s) - Q_{\phi}(s, a) + \log \pi_{\phi}(a | s))$$

7. Train Q to minimize

$$J_Q(\theta) = \mathbb{E}_{(s, a) \sim \rho} (\frac{1}{2} (Q_{\theta}(s, a) - \hat{Q}(s, a))^2)$$

a. where

$$\hat{Q}(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim p} (V_{\phi}(s'))$$

b. one point estimator:

$$\hat{S} J_Q(\theta) = \nabla_{\theta} Q_{\theta}(a, s) (Q_{\theta}(s, a) - r(s, a) - \gamma V_{\phi}(s'))$$

c. $\bar{\phi}$ - target network param

8. last, pick θ to ϕ to minimize

$$J_{\pi}(\theta) = \mathbb{E}_{s \sim \rho} (D_{KL}(\pi_{\theta}(\cdot | s) \| \frac{\exp(Q_{\theta}(s, \cdot))}{Z_{\theta}(s)}))$$

9. now, let's actually choose

$$a_t = f_{\theta}(x_t, s_t)$$

89 a. θ is not, e.g. from Gaussian

b. this implicitly defines θ in terms of f_{θ}

10. then we find we may as well minimize

$$J_{\pi}(\theta) = \mathbb{E}_{s \sim p, \pi} (\log \pi(a_t | s_t) - Q_{\theta}(s_t, f_{\theta}(x_t, s_t)))$$

a one point estimator:

$$\mathbb{E} J_{\pi}(\theta) = \mathbb{E} \log \pi(a_t | s_t)$$

$$+ (\mathbb{E} \log \pi(a_t | s_t) - \mathbb{E} Q_{\theta}(s_t, a_t)) \mathbb{E} f_{\theta}(x_t, s_t)$$

11. finally, we actually find two Q_{θ} 's, $\uparrow$
we then min. here

12. Q can be a replay buffer - it's all policy.